UNIT I: Vehicle Structure & Engine Components

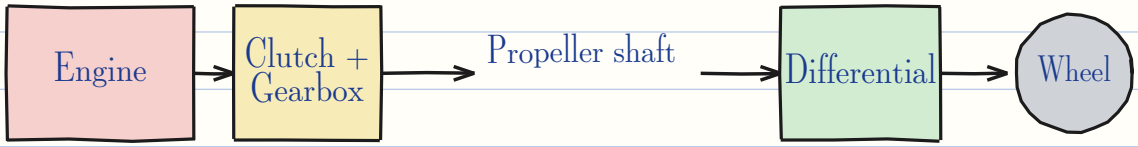
Syllabus: Vehicle construction, chassis & body, specifications, engine types, location, cylinder arrangement, block, head, liners, piston, rings, pin, connecting rod, crankshaft, valves, lubrication, oil pumps, filters, crankcase ventilation.

1. Vehicle Construction

- An automobile is a self-propelled road vehicle that carries passengers or goods, powered by its own engine.
- Basic parts of any vehicle: Chassis + Body (together they make the complete vehicle).
- Chassis = whole vehicle except the body, i.e. everything needed to make it run (a "running chassis").

Main components of the chassis

- Frame - the backbone that carries all the loads.
- Engine - power source (petrol / diesel / hybrid / electric motor).
- Transmission - clutch, gearbox, propeller shaft, final drive, differential, axles.
- Suspension - springs + shock absorbers; isolates road shocks.
- Steering system - controls direction of travel.
- Brakes - slow down / stop / park the vehicle.
- Wheels and tyres - support weight, give traction.
- Electrical system - battery, alternator, starter, lights, ignition.



Power flow in a front-engine rear-wheel-drive (FR) chassis

Fig 1.1 Power flow: engine -> wheels

Remember

Complete vehicle = chassis + body. Chassis without body is called a "chassis cab" or rolling chassis; used for trucks and buses where body is built by a body-builder.

2. Chassis, Body & Specifications

Frame (chassis frame) - functions

- Supports the engine, transmission, body and passengers; carries static + dynamic loads.
- Resists torsion, bending (vertical loads) and side thrust / braking forces.

Types of frame / construction

1. Conventional (ladder) frame - two long side members + cross members. Used in trucks, buses, jeeps. Body is bolted on separately; heavy but strong.
2. Integral / unitary (monocoque) frame - no separate frame; body panels + floor pan form one stressed structure. Used in cars. Light, low floor, less cost; harder to repair.
3. Semi-integral frame - partial frame (sub-frame) at front carries engine and suspension, rest is integral body.

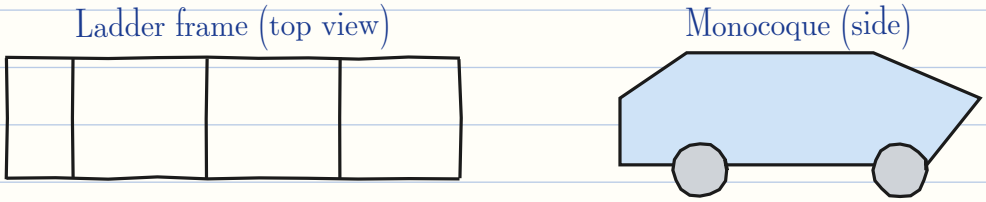


Fig 1.2 Ladder frame and unitary body

Body - functions & types

- Provides passenger / luggage space, protection, aerodynamics and appearance.
- Car body types: saloon (sedan), hatchback, estate, coupe, convertible, SUV, MPV.
- Commercial: truck (platform, tipper, tanker), bus (single / double decker), van.

Vehicle specifications (typical data sheet)

Item	Typical entry
Engine	Type, no. of cylinders, bore x stroke, capacity (cc), CR
Power / Torque	kW (bhp) @ rpm ; Nm @ rpm
Transmission	5-speed manual / AMT / CVT / AT
Dimensions	Length, width, height, wheelbase, track, ground clearance
Weights / others	Kerb weight, GVW, tyre size, fuel tank, max speed

3. Engine - Types

Engine = a machine that converts heat energy of fuel into mechanical work. In vehicles we mainly use Internal Combustion (IC) engines - combustion takes place inside the cylinder.

Classification of engines

Basis	Types
Type of combustion	IC engine, EC engine (steam engine, Stirling)
Fuel used	Petrol (SI), Diesel (CI), Gas / CNG / LPG, bi-fuel, ethanol
Ignition	Spark Ignition (SI) , Compression Ignition (CI)
Cycle of operation	Otto (SI), Diesel (CI), Dual
No. of strokes	2-stroke, 4-stroke
Cooling	Air-cooled, water (liquid) cooled
Cylinder arrangement	In-line, V, flat (boxer), radial, W
Valve arrangement	L, I, F, T head - OHV, OHC (SOHC / DOHC)

Valve arrangements (head types)

- L-head (side valve) - both valves on one side of the cylinder; old engines.
- I-head (overhead valve, OHV) - valves in cylinder head, operated by push rods & rockers.
- OHC - camshaft in the head: SOHC (one cam) or DOHC (two cams, 4 valves/cyl). Best breathing at high rpm.

Terms used in engines

TDC / BDC = top / bottom dead centre. Stroke L = distance TDC to BDC.
Bore D = cylinder diameter.
Swept volume Vs = (pi/4) D^2 L. Compression ratio r = (Vs + Vc) / Vc (Vc = clearance volume).

4. Four-Stroke Engine Working

One cycle = 4 strokes = 2 crankshaft revolutions (720 deg). Only 1 power stroke per 2 revolutions. Valves: inlet (IV) and exhaust (EV).

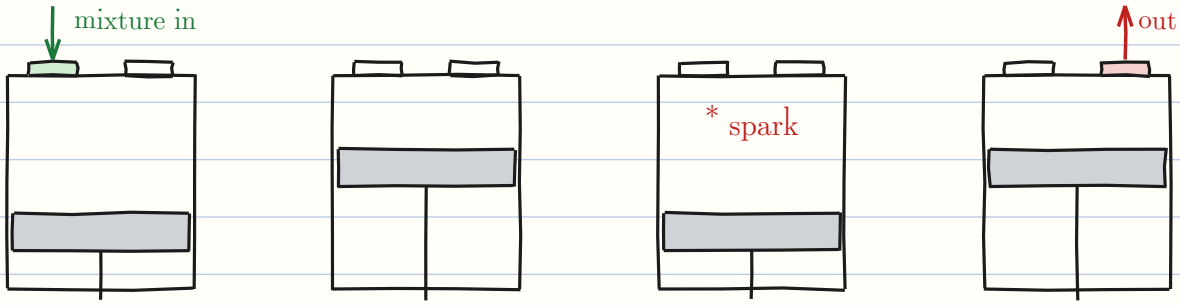


Fig 1.3 Four strokes of the Otto cycle (SI engine)

Strokes

1. Suction: piston TDC -> BDC; IV open, EV closed; fresh air-fuel mixture (petrol) / only air (diesel) is drawn in.
2. Compression: BDC -> TDC; both valves closed; charge compressed (CR 7-11 petrol, 16-22 diesel); temperature and pressure rise.
3. Power (expansion): spark (petrol) / fuel injection & self-ignition (diesel) near TDC; burnt gases push piston TDC -> BDC; work is produced.
4. Exhaust: BDC -> TDC; EV open, IV closed; burnt gases pushed out.

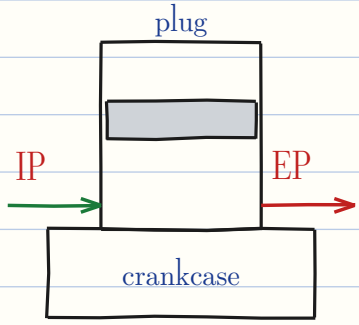
- Flywheel stores energy of power stroke to carry the engine through the other 3 (idle) strokes.

Petrol vs Diesel (4-stroke)

Petrol: mixture drawn, spark plug ignites, CR 6-10, Otto cycle. Diesel: only air drawn, fuel injected at end of compression, self-ignition, CR 16-22, Diesel cycle. Diesel engines give higher thermal efficiency (~35-40 %).

5. Two-Stroke Engine & Comparison

- Two-stroke engine completes a cycle in 2 strokes = 1 revolution of crankshaft (360 deg). Power stroke every revolution.
- No valves - ports (inlet, transfer, exhaust) are opened and closed by the piston itself.
- Crankcase-scavenged: fresh charge is compressed in crankcase and transferred into cylinder.



Upward stroke: compression above piston, suction (fresh charge entering) below piston.

Downward stroke: power above piston, crankcase compression, then transfer port opens; exhaust port opens first.

IP = inlet port, EP = exhaust port (TP = transfer port)

Fig 1.4 Crankcase scavenged 2-stroke petrol engine

Comparison: 2-stroke vs 4-stroke

Point	2-stroke	4-stroke
Power stroke	Once per revolution	Once per 2 revolutions
Valves	Ports (no valves)	Inlet + exhaust valves
Power for same size	Higher (theoretically 2x)	Lower
Fuel economy / emissions	Poor, more HC (oil mixed)	Better, cleaner
Lubrication	Petroil (oil mixed in fuel)	Separate oil sump
Use	Scooters (old), chainsaw, outboard	Cars, trucks, bikes

Scavenging

Process of pushing out burnt gases with fresh charge. Types: cross-flow, loop (Schnurle), uniflow. Loss of fresh mixture with exhaust (short-circuiting) is a drawback of 2-stroke petrol engines.

6. Location of Engine & Drive Layouts

The position of the engine and driven wheels decides interior space, weight distribution, traction and handling.

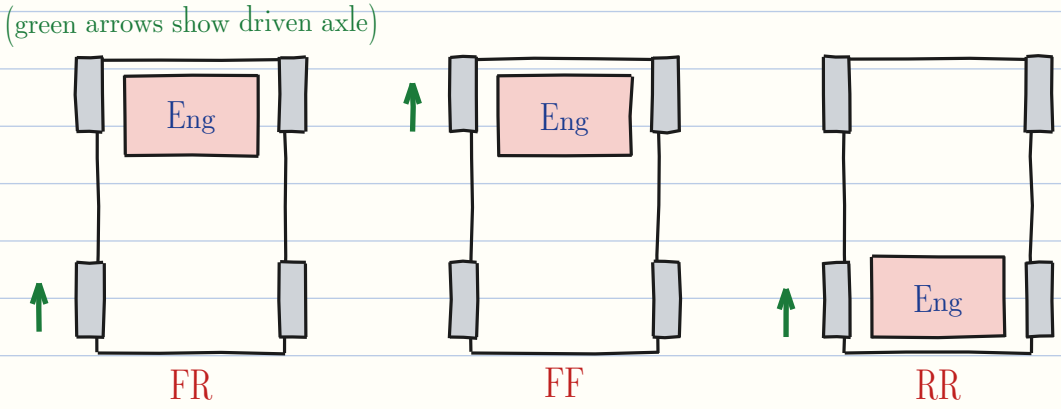


Fig 1.5 FR, FF and RR layouts (top view)

Layouts

- Front engine - rear wheel drive (FR): conventional. Needs propeller shaft; good weight distribution, better traction on hills; used in trucks, SUVs, sports cars.
- Front engine - front wheel drive (FF): compact, no propeller shaft or tunnel, more cabin space, good stability; most modern small cars. Disadvantage: understeer, complex CV joints.
- Rear engine - rear wheel drive (RR): engine over drive wheels -> good traction; used in Tata Nano, VW Beetle, Porsche 911, some buses. Disadvantage: oversteer, cooling problems.
- Mid-engine (MR): engine in between axles, best balance; sports/race cars (Ferrari, Lamborghini).
- All-wheel / 4-wheel drive (AWD/4WD): power to all wheels via transfer case - off-roaders.

Advantage of front engine

Better engine cooling (air flow), simple controls, large luggage space, easy service.

7. Cylinder Arrangement

Number and arrangement of cylinders affects balance, smoothness, size and power. Multi-cylinder engines give smoother torque than a single cylinder.

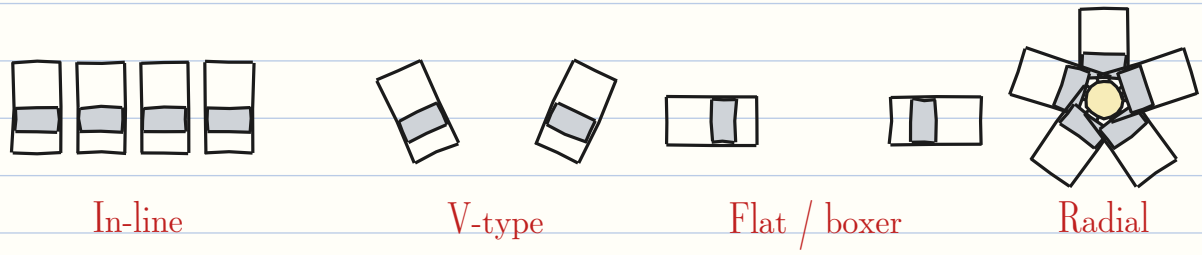


Fig 1.6 Common cylinder arrangements

- In-line: all cylinders in a single row; 3, 4, 5, 6 cyl.; simple & cheap, long engine. Inline-4 most common.
- V-type: two banks at 60-90 deg; short and rigid; V6, V8, V12 - used in luxury / high power cars.
- Flat / horizontally-opposed (boxer): low centre of gravity, perfectly balanced; Subaru, Porsche.
- Radial: cylinders around a crankcase like a star; aircraft engines.
- W-type: three banks (VW W12 / Bugatti W16); Slant engines tilted to lower the bonnet.

Firing order

- Sequence in which power strokes occur in cylinders. Chosen to get even torque, good balance, minimum vibration, uniform cooling.

Engine	Firing order
4-cylinder in-line	1 - 3 - 4 - 2 or 1 - 2 - 4 - 3
6-cylinder in-line	1 - 5 - 3 - 6 - 2 - 4
V8	1 - 8 - 4 - 3 - 6 - 5 - 7 - 2
3-cylinder	1 - 2 - 3

- Firing interval for a 4-stroke = 720 / number of cylinders (4 cyl -> 180 deg; 6 cyl -> 120 deg).

8. Cylinder Block & Crankcase

- The cylinder block is the main structure of the engine - it holds cylinders, water jackets, oil galleries and supports crankshaft, head and accessories.
- Upper part contains the cylinders (bores); lower part is the crankcase which houses crankshaft & main bearings; the oil pan (sump) is bolted below it.

Materials

- Grey cast iron (with Ni, Cr) - good wear resistance, damping, low cost; heavier.
- Aluminium alloy - light (up to 50 % less), good heat transfer; needs cast-iron liners or special bore coating (Nikasil).

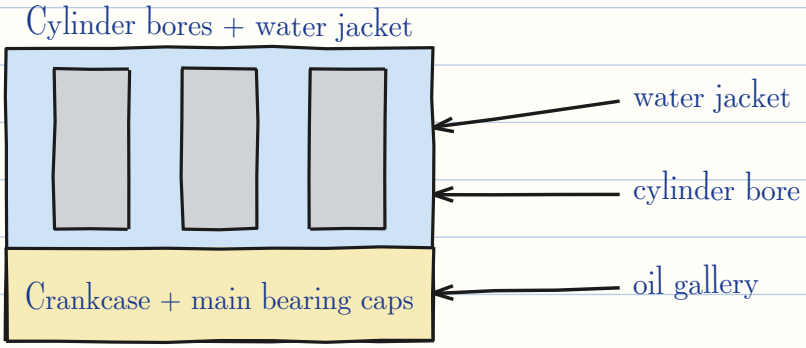


Fig 1.7 Sectional view of a cylinder block

Features of the block

- Water jackets around cylinders (liquid cooling) or fins (air cooling).
- Oil galleries / drilled passages to feed crank bearings, cam bearings and head.
- Deck surface machined flat for head gasket; tappet bores, motor mounting lugs.
- Open deck (jacket open at top, easy casting) vs closed deck (stronger; better for turbo).
- Cylinder bores may be cast in block ("integral") or fitted with liners.

Requirements

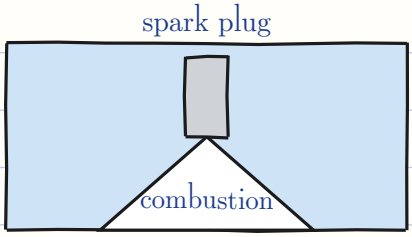
High strength & rigidity, good thermal conductivity, minimum distortion, easy casting & machining, resistance to wear and corrosion.

9. Cylinder Head & Head Gasket

- Cylinder head is bolted on top of the block; it closes the cylinders and forms the combustion chamber.
- Carries inlet & exhaust valves, ports, spark plug (SI) / injector (CI), valve seats, guides, rocker arms or camshafts, water & oil passages.
- Material: cast iron (old) or aluminium alloy (modern - lighter, cooler running).

Combustion chamber shapes (SI)

- Wedge, hemispherical (crossflow, spark plug central - best), bath-tub, pent-roof (4 valve), Heron / bowl-in-piston.
- Good chamber = high compression ratio without knock, short flame travel, turbulence, compact (low surface/volume ratio).



Hemispherical chamber
- valves inclined, large valve area
- central plug, short flame path

Fig 1.8 Hemispherical combustion chamber

Head gasket

- Thin sheet placed between block and head to seal compression gases, coolant and oil.
- Types: copper-asbestos (old), steel-asbestos, multi-layer steel (MLS), composite graphite.
- Head bolts are tightened in a specified sequence (spiral from centre outwards) to the correct torque, in 2-3 stages.
- Blown head gasket symptoms: white smoke, overheating, coolant in oil (milky), loss of compression.

Diesel head

Contains pre-combustion chamber / swirl chamber (IDI) or injector directly over piston bowl (DI); glow plug for cold starting.

10. Cylinder Liners

- A liner (sleeve) is a cylindrical lining inserted into the block; the piston rings slide on its inner surface.
- Why use liners? (1) Block can be made of light material, (2) liner can be hard wear-resistant, (3) worn liner is replaced instead of the whole block, (4) easy repair / reboring.
- Material: alloy cast iron (Ni, Cr, Mo), nitrided steel; inner surface honed (cross-hatch pattern) to hold oil film.

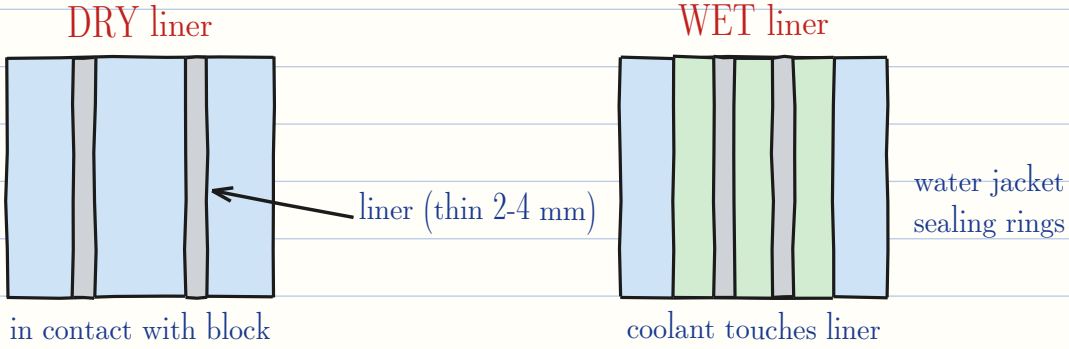


Fig 1.9 Dry and wet cylinder liners

	Dry liner	Wet liner
Contact with coolant	No (pressed into block bore)	Yes - outer surface is cooled directly
Thickness	Thin (1.5 - 4 mm)	Thick (5 - 8 mm)
Cooling	Poorer	Better
Sealing	No sealing problem	Needs rubber rings / flange to stop leakage
Replacement	Needs press / machine	Easy to remove & fit
Use	Light-duty petrol engines	Heavy-duty diesel engines

- Honing, chrome plating, nitriding of liner bore improve wear life. Wear is greatest near TDC (top ring reversal) -> ridge forms.
- Aluminium blocks use cast-in liners or Nikasil / plasma-sprayed coatings.

11. Piston

- The piston moves in the cylinder; it receives gas force and transmits it to the connecting rod through the piston pin; seals the cylinder with rings.

Piston parts

- Crown / head - top surface; flat, domed, or with bowl (diesel).
- Ring belt / lands & grooves - hold compression rings and oil control ring.
- Skirt - guides piston, takes side thrust. Boss - holds gudgeon pin. Thrust faces - minor / major thrust sides.

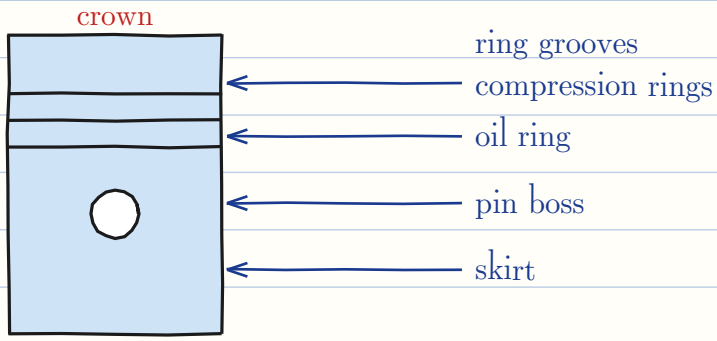


Fig 1.10 Piston (sectional side view)

Materials & design

- Aluminium alloy (Al-Si, 12 % Si "Y-alloy / Lo-Ex") - light, high conductivity; cast or forged. Cast iron pistons only in slow diesels.
- Aluminium expands more than cast iron -> running clearance 0.03 - 0.1 mm; tapered / elliptical (cam-ground) skirt, crown narrower as it is hotter.
- Slots (T-slot, U-slot) give flexibility and heat barrier; steel struts (autothermic) control expansion.
- Skirt types: slipper skirt (cut-outs, light), split skirt, solid skirt, offset pin to reduce piston slap.

Piston slap

Knocking noise from excess clearance when piston changes side-thrust direction at TDC - reduced by offsetting the pin, tapered skirt, correct clearance.

12. Piston Rings & Piston Pin

Piston rings

- Split rings fitted in grooves; spring outward to press against cylinder wall.
- Functions: (1) seal combustion gas (compression), (2) control oil film (scraping), (3) transfer heat from piston to the cylinder wall.

Types

- Compression rings (top 2) - barrel-faced / taper-faced, chrome plated; top ring takes highest heat & wear.
- Oil control (scraper) ring (bottom) - slotted, with expander spring; returns excess oil to sump through holes in groove.

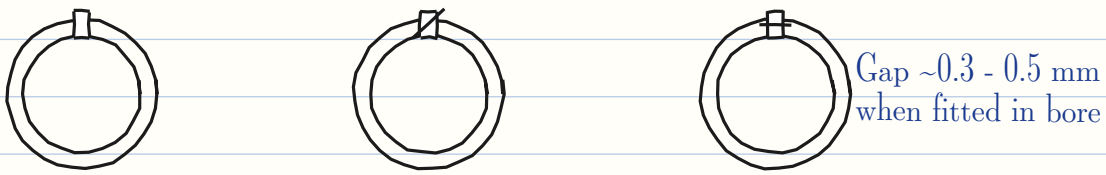


Fig 1.11 End-gap (joint) types

- Ring materials: grey cast iron (alloyed), steel with chrome / molybdenum coating. Rings are staggered 120 deg apart when fitting.
- Faults: ring sticking (carbon), ring wear -> blow-by, high oil consumption, blue smoke.

Piston pin (gudgeon pin / wrist pin)

- Hollow, case-hardened steel pin connecting piston and small end of connecting rod; carries the gas load.
- Hollow = lighter, reduces inertia, strong in bending & shear; ground and polished surface.

Type	Arrangement
Fixed (in piston)	Pin clamped by bolt in boss; free to turn in rod small-end bush
Semi-floating	Pin fixed to rod (clamp), free to turn in piston bosses
Full-floating	Free in both piston & rod; held by circlips at ends - most common

13. Connecting Rod

- Links piston and crankshaft; converts the reciprocating motion of piston into rotary motion of crank and transmits the force.
- Parts: small end (piston pin, bronze bush), shank (I-section), big end (split, two bearing shells with bolts & cap).

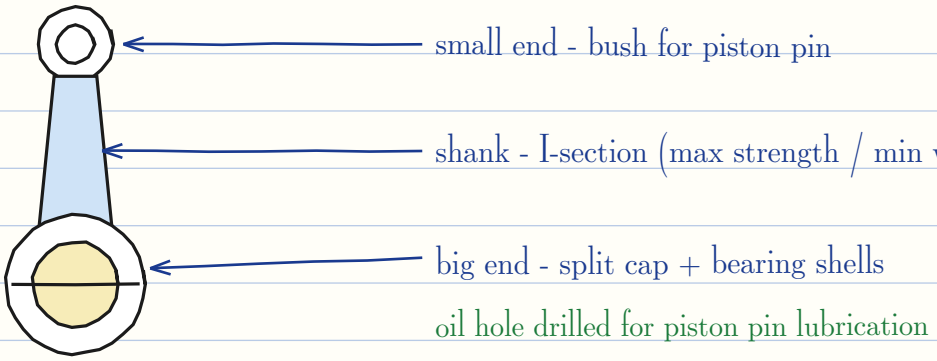


Fig 1.12 Connecting rod

Materials

- Carbon steel (0.4 %C), nickel / chromium-molybdenum alloy steel forged (drop-forged), aluminium (small / racing), titanium (racing), powder metal / cracked-cap rods.

Design points

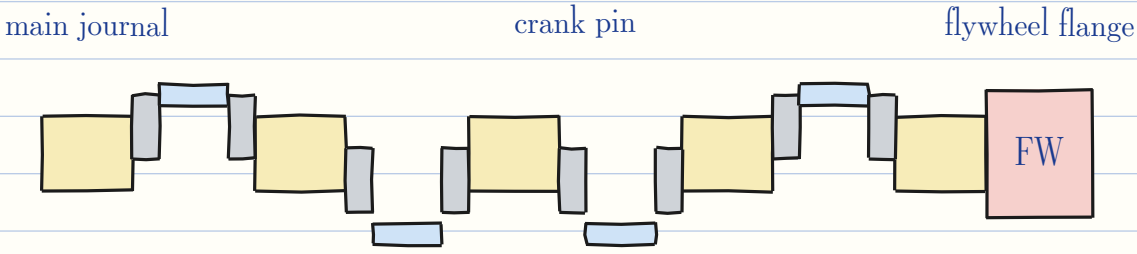
- I-section best: high bending strength, min. weight (small inertia); H-section also used in racing engines.
- Length of rod / crank radius ratio (l/r = 3.5 - 4.5) - longer rod reduces side thrust on piston but increases engine height.
- Big end bolts under high tension; made of high tensile steel and self-locking nuts.
- Bearing shell: steel backing + lining of babbitt / copper-lead / aluminium-tin, with locating tang; thickness uniform.
- Rods are weight-matched per cylinder to minimize vibration; marks on the cap and rod must match (numbered).

Stresses on connecting rod

Compression (gas pressure), tension (inertia at TDC), bending due to whirling (buckling). Designed as strut using Euler / Rankine formula.

14. Crankshaft

- The crankshaft converts reciprocating motion of pistons into rotary motion and delivers torque to the flywheel & transmission.
- Parts: main journals (run in main bearings), crank pins (big end journals), crank webs / arms, counterweights, fillets, oil holes, flange (flywheel), nose (pulley / timing gear).



webs + counterweights balance the rotating & reciprocating masses

Fig 1.13 Four-cylinder crankshaft (schematic)

Materials & manufacture

- Forged alloy steel (Ni-Cr-Mo / EN 8, EN 24 / 40Cr) - high strength; cast nodular (SG) iron - cheaper, good damping (cars).
- Journals are induction hardened / nitrided, ground and polished; fillets rolled to avoid fatigue cracks.

Features

- Number of main bearings: n+1 (for n cyl) = strongest; 3 for 4-cyl is common (economy).
- Crank throw arrangement: 4-cyl in-line -> 180 deg; 6-cyl in-line -> 120 deg; V8 -> 90 deg (cross-plane).
- Torsional vibration damper (harmonic balancer) on nose end absorbs twisting vibrations.
- Flywheel at the rear stores energy & smooths torque; carries ring gear for starter and clutch face.

Crankshaft failure

Fatigue crack at fillets, bearing seizure (lack of oil), bent shaft; check run-out & journal ovality / taper with micrometer.

15. Valves & Valve Train

- Poppet (mushroom) valves open and close the intake & exhaust ports at proper timing; seal the combustion chamber during compression & power.

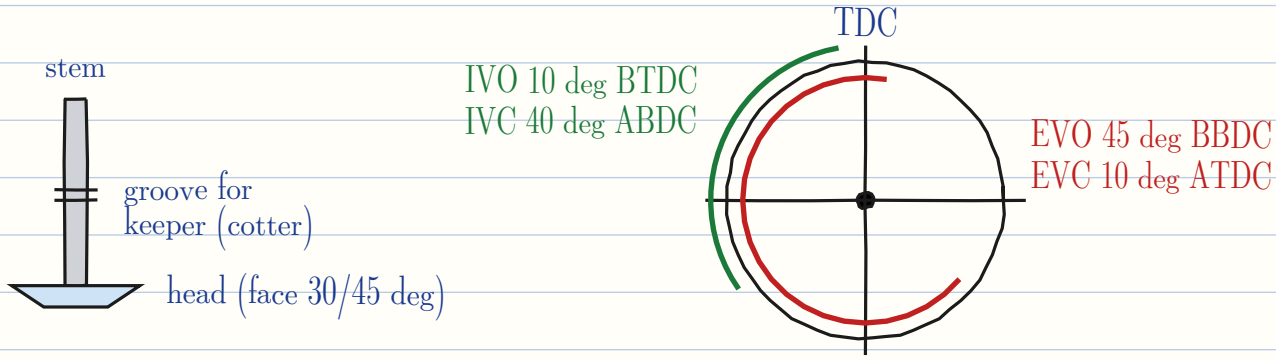


Fig 1.14 Poppet valve and valve timing diagram

Valve materials

- Inlet: chrome / nickel / silicon-chrome steel. Exhaust (runs at 700-800 deg C): austenitic steel (21-4N), Nimonic; stellite-faced seat; sodium-filled hollow stems for cooling.

Parts of valve mechanism

- Camshaft (cam lobes) -> tappet / cam follower -> push rod -> rocker arm -> valve; valve spring (+ retainer & keepers) closes valve; valve seat & valve guide in head.
- Valve clearance (tappet clearance) 0.2-0.3 mm cold; compensates thermal expansion. Hydraulic lifters give zero-lash automatically.
- Camshaft drive: gear, chain or toothed (timing) belt; runs at half crankshaft speed (4-stroke).
- Valve timing: inlet opens before TDC, closes after BDC; exhaust opens before BDC, closes after TDC; overlap period improves breathing. VVT / VTEC alters timing with rpm.
- Faults: valve burning, sticking, spring fatigue, seat recession, wrong tappet gap.

16. Lubrication System

- Lubrication = supplying oil between moving surfaces to form a film that reduces friction & wear.

Functions of lubricating oil

- 1) Reduces friction & wear 2) Cools parts (piston, bearings) 3) Seals piston rings 4) Cleans (carries away carbon, metal particles) 5) Protects against corrosion 6) Damps noise / shock loads.

Types of lubrication systems

1. Petroil / mist system - oil mixed with petrol (2-stroke engines, 20:1 - 50:1).
2. Splash system - dippers on big ends splash oil from troughs; used in small stationary / lawn-mower engines.
3. Pressure feed (full pressure) system - oil pump sends oil under 2-4 bar to all main bearings, big ends, cam bearings; most cars.
4. Semi-pressure - pressure to main bearings, splash for cylinder walls.
5. Dry sump - oil stored in separate tank; scavenge pump returns oil; used in racing / sports cars & aircraft (no oil starvation in corners).
6. Wet sump - oil kept in the sump below crankcase; simple; most vehicles.

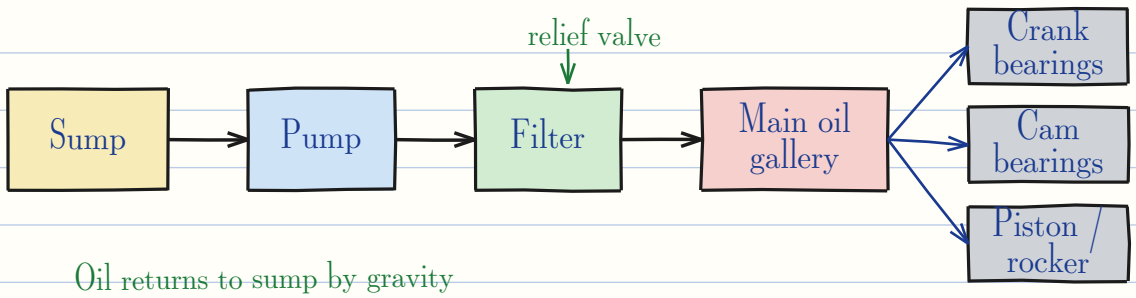


Fig 1.15 Wet-sump pressure feed lubrication

- Oil pressure relief valve keeps pressure constant (~3 bar). Oil pressure warning lamp / gauge on dash.
- Oil grade: SAE viscosity (e.g. 20W-40, 5W-30); API service class SL, SN (petrol), CI-4 (diesel). Viscosity index is high for multigrade oils.

17. Oil Pumps, Filters & Crankcase Ventilation

Oil pumps

- Gear pump: two meshed gears (drive + driven) trap oil between teeth and housing; simple, robust, driven from camshaft / crankshaft.
- Rotor (gerotor) pump: inner rotor with n lobes inside outer rotor with n+1 lobes; compact, quiet, used on the crank nose.
- Vane pump - sliding vanes in eccentric rotor, variable-displacement versions save energy.

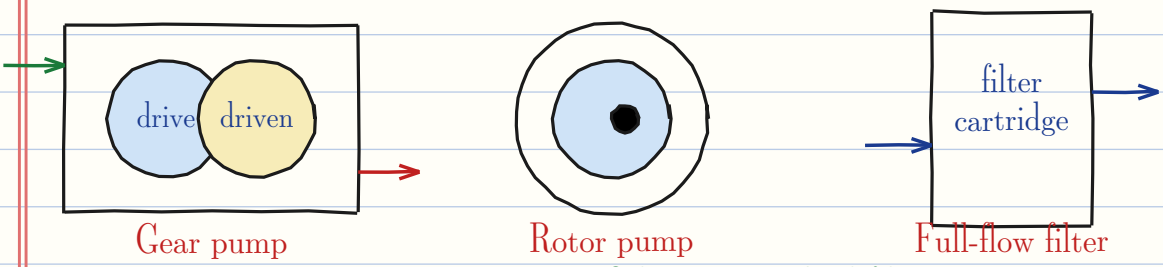


Fig 1.16 Oil pumps and oil filter

Oil filters

- Remove carbon, metal dust, sludge. Element: paper (pleated), felt, cotton waste, metal-mesh. Strainer (wire mesh) on pump pick-up in sump.
- Full-flow filter - all oil passes through; a bypass valve opens if filter clogs. Bypass filter - only 10 % oil filtered continuously (finer). Change every 5000-10000 km.

Crankcase ventilation

- Blow-by gases (unburnt HC, water vapour, acids) leak past rings into the crankcase -> dilute / contaminate oil, cause sludge, corrosion & pressure build-up.
- Open (road-draught) ventilation: breather tube vents fumes to atmosphere - pollution (old).
- Closed / PCV (Positive Crankcase Ventilation): fumes drawn into the intake manifold through the PCV valve and burnt; fresh air enters through the breather / air filter. Reduces HC emission.

PCV valve

Spring-loaded; at idle (high vacuum) it restricts flow, at full load (low vacuum) opens fully. A stuck valve causes rough idle, oil leaks, sludge.